

Let's next focus on a provider network, say AS B. Suppose that B has learned (from A) that A has a path AW to W. B can thus install the route BAW into its routing information base. Clearly, B also wants to advertise the path BAW to its customer, X, so that X knows that it can route to W via B. But should B advertise the path BAW to C? If it does so, then C could route traffic to W via CBAW. If A, B, and C are all backbone providers, then B might rightly feel that it should not have to shoulder the burden (and cost!) of carrying transit traffic between A and C. B might rightly feel that it is A's and C's job (and cost!) to make sure that C can route to/from A's customers via a direct connection between A and C. There are currently no official standards that govern how backbone ISPs route among themselves. However, a rule of thumb followed by commercial ISPs is that any traffic flowing across an ISP's backbone network must have either a source or a destination (or both) in a network that is a customer of that ISP; otherwise the traffic would be getting a free ride on the ISP's network. Individual peering agreements (that would govern questions such as those raised above) are typically negotiated between pairs of ISPs and are often confidential; [Huston 1999a] provides an interesting discussion of peering agreements. For a detailed description of how routing policy reflects commercial relationships among ISPs, see [Gao 2001; Dimitropoulos 2007]. For a discussion of BGP routing policies from an ISP standpoint, see [Caesar 2005b].

As noted above, BGP is the *de facto* standard for inter-AS routing for the public Internet. To see the contents of various BGP routing tables (large!) extracted from routers in tier-1 ISPs, see <http://www.routeviews.org>. BGP routing tables often contain tens of thousands of prefixes and corresponding attributes. Statistics about the size and characteristics of BGP routing tables are presented in [Potaroo 2012].

This completes our brief introduction to BGP. Understanding BGP is important because it plays a central role in the Internet. We encourage you to see the references [Griffin 2012; Stewart 1999; Labovitz 1997; Halabi 2000; Huitema 1998; Gao 2001; Feamster 2004; Caesar 2005b; Li 2007] to learn more about BGP.

4.7 Broadcast and Multicast Routing

Thus far in this chapter, our focus has been on routing protocols that support unicast (i.e., point-to-point) communication, in which a single source node sends a packet to a single destination node. In this section, we turn our attention to broadcast and multicast routing protocols. In **broadcast routing**, the network layer provides a service of delivering a packet sent from a source node to all other nodes in the network; **multicast routing** enables a single source node to send a copy of a packet to a subset of the other network nodes. In Section 4.7.1 we'll consider broadcast routing algorithms and their embodiment in routing protocols. We'll examine multicast routing in Section 4.7.2.

4.7.1 Broadcast Routing Algorithms

Perhaps the most straightforward way to accomplish broadcast communication is for the sending node to send a separate copy of the packet to each destination, as shown in Figure 4.43(a). Given N destination nodes, the source node simply makes N copies of the packet, addresses each copy to a different destination, and then transmits the N copies to the N destinations using unicast routing. This **N-way-unicast** approach to broadcasting is simple—no new network-layer routing protocol, packet-duplication, or forwarding functionality is needed. There are, however, several drawbacks to this approach. The first drawback is its inefficiency. If the source node is connected to the rest of the network via a single link, then N separate copies of the (same) packet will traverse this single link. It would clearly be more efficient to send only a single copy of a packet over this first hop and then have the node at the other end of the first hop make and forward any additional needed copies. That is, it would be more efficient for the network nodes themselves (rather than just the source node) to create duplicate copies of a packet. For example, in Figure 4.43(b), only a single copy of a packet traverses the R1-R2 link. That packet is then duplicated at R2, with a single copy being sent over links R2-R3 and R2-R4.

The additional drawbacks of N -way-unicast are perhaps more subtle, but no less important. An implicit assumption of N -way-unicast is that broadcast recipients, and their addresses, are known to the sender. But how is this information obtained? Most likely, additional protocol mechanisms (such as a broadcast membership or destination-registration protocol) would be required. This would add more overhead and, importantly, additional complexity to a protocol that had initially seemed quite simple. A final drawback of N -way-unicast relates to the purposes for which broadcast is to be used. In Section 4.5, we learned that link-state routing protocols use broadcast to disseminate the link-state information that is used to compute unicast routes. Clearly, in situations where broadcast is used to create and update unicast routes, it would be unwise (at best!) to rely on the unicast routing infrastructure to achieve broadcast.

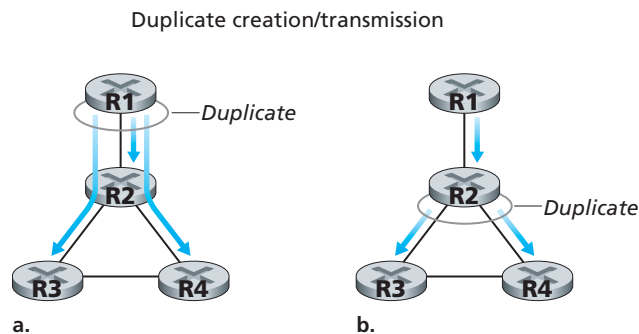


Figure 4.43 ♦ Source-duplication versus in-network duplication